Evaluating Predictive Power of Renewable and Non-renewable Energy market indices in forecasting the Artificial Intelligence market index in the United States

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Abstract

This paper evaluates the predictive power of renewable and non-renewable energy indices in forecasting the AI Index in the US. It employs two approaches: a Structural Vector Auto-Regression (SVAR) Model and Machine Learning Models like Support Vector Regression (SVR) and Extreme Gradient Boosting Regression (XGBoost). The primary findings in this paper were that shocks in both the energy indices had a significant immediate impact on the AI index, thus suggesting the presence of a contemporaneous relationship between the two. Furthermore, the Support Vector Regression with technical indicators, along with the additional lagged terms relative to the SVAR model, had the lowest RMSE and MAE across all sets of models. Lastly, all the Machine Learning models performed significantly better on out-of-sample data compared to the SVAR model.

JEL Codes: G17, C53, C58, C22

1. Introduction and Literature Review

Since the advent of Large Language Models (LLMs), numerous pivotal fields have adopted these technologies in their operations, thus contributing to the spread of Artificial Intelligence Technologies and, in turn, resulting in the recent AI boom. These fields include research in drug discovery, medicine, and especially finance through the development of quantitative models, which has now become significantly easier due to the presence of these models (Wang, Chu, Doan et al. 2024). However, these AI infrastructures are primarily dependent on data centers, which sustain these tools using as much electricity as that of 100,000 households annually in the US (IEA 2025). Furthermore, this production of electricity is mainly dependent on coal (about 30%), followed by renewable sources such as solar, hydro, and wind energy (about 27%) (IEA 2025). Also, with the increasing influence of AI, a lot of workplaces have started incorporating it into their daily operations for technological advancements and efficiency increases.

This continuous progression of AI requires expansion of data centers and eventually a massive increase in energy demand, which would in turn be reflected in their energy producers' stock prices through better performance and a surge in consumption. Additionally, this change in the price of energy firms' stocks could serve as a crucial indicator for policymakers to stabilize the economy before innovation breakthroughs that cause financial volatility in the AI sector, or for investors with portfolios centered on such AI companies. This heavy dependence of AI on both renewable and non-renewable sources points to the connectedness of changes in the demand for energy and the supply of Artificial Intelligence infrastructure, and the performance of such companies. This relationship shapes my motivation in identifying whether the stock performance of these energy-producing firms contains predictive power in determining the changes in stock performance of publicly traded AI giants in the US.

Thus, my research question is to evaluate the predictive power of energy-producing firms' (renewable and non-renewable) stock performance in forecasting the performance of Artificial Intelligence (AI) companies in the US. This paper addresses this question through analysis utilizing two main modeling approaches: the SVAR (Structural Vector Auto Regression) model and Machine Learning algorithms. Both approaches use the Clean Energy and Oil and Gas sector indices to predict the corresponding AI stock price index. Lastly, this analysis is followed by an evaluation of these model predictions using out-of-sample testing data using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) as evaluation metrics.

This paper uses the NASDAQ Clean Energy Index (NASDAQ 2026b) and the Dow Jones U.S. Oil and Gas Index (S&P Global 2026) as predictors to determine the NASDAQ CTA Artificial Intelligence and Robotics Index (NASDAQ 2026a). This choice of the AI index was inspired from previous researchers like Urom, Ndubuisi, Mzoughi et al. (2024). In addition to this, macroeconomic indicators like the S&P 500 Global Index (S&P Global 2025) and the Effective Federal Funds Rate (FRED 2026) are also included to prevent any omitted variable bias in these results. All these variables span from 2021 to 2025, and are recorded at a daily frequency.

The primary model used for analysis is a Structural Vector Auto-Regression (SVAR) with a short-run restriction. This model assumes the ordering to be as follows: Oil and Gas Index, Clean Energy Index, AI Index, S&P 500 Index, and lastly the Effective Federal Funds Rate. Additionally, this paper compared the forecast metrics on out-of-sample data across both the SVAR model and Machine Learning (ML) models like Support Vector Regression and Gradient Boosting Regression. For the ML models, two sets of features are used: one set of input features is essentially identical to the input in the SVAR model's corresponding equation for the AI Index, the second set of features includes additional lagged terms for all features along with certain technical indicators like the Relative Strength Index (RSI), and Simple and Exponential Moving Averages (SMA, EMA) to capture the momentum in the AI index trends. The use of these indicators was inspired by previous studies like Agrawal, Khan, and Shukla (2019).

Some important findings in this research are briefly discussed below. First, the Machine Learning models using either set of features outperform the SVAR model in terms of prediction accuracy by an out-of-sample RMSE of almost 300 units. Out of the Machine Learning models, the Support Vector Regression with technical indicators and additional lagged terms performs the best in predicting the AI Index with a negligible RMSE of 13.767 units. Second, based on the impulse response functions and the forecast error variance decomposition, it is evident that shocks in the Clean Energy Index appear to have a significant positive impact on the NASDAQ AI Index when assumed to have a contemporaneous relationship. Lastly, shocks in the Dow Jones U.S. Oil and Gas Index also significantly affect the NASDAQ AI Index, irrespective of the ordering assumption used in the paper, which suggests the energy-AI relationship for non-renewable sources is still evident despite the efforts being made for an energy transition in contemporary times.

The energy-AI relationship has been an important branch of research over the last few years, with multiple researchers contributing to this field. Researchers have found that clean energy stock prices are affected by stock prices of both technology firms and oil prices and by their past movements(Bondia, Ghosh, and Kanjilal 2016; Kumar, Managi, and Matsuda 2012). Both these studies focus on technology firms instead of specifically AI firms, as these studies were published before the onset of Large Language Models, which took off in 2020 (Wang, Chu, Doan et al. 2024). This research focuses extensively on the Artificial Intelligence Index and how it is impacted by energy indices, which extends this existing branch of literature involving energy-technology dynamics from just technology-based firms to Artificial Intelligence firms.

Liu, Huang, and Liu (2026) found the existence of a strong correlation between Artificial Intelligence and Clean Energy, which reflects investment interdependence between these industries. Similarly, Raggad and Bouri (2025) established that the connectedness between returns from clean energy and AI is more than that of dirty energy and AI. Additional research focused on the causal relationship and directional predictability of the returns on AI stocks and the energy sector using methods like wavelet coherence analysis, suggesting significant predictive power of energy-sector returns in determining AI stock returns (Urom, Ndubuisi, Mzoughi et al. 2024). Further research focusing on inter-connectedness of volatility and frequency among AI stocks, bonds, and fossil fuel markets shows that there is a strong connection between these markets, which might cause spillovers of effects on an individual market (Yousaf, Ijaz, Umar et al. 2024). These research studies focus on the causal relationship between the energy and AI markets as well as on the connectedness of these markets in terms of volatility and spillovers of effects between markets. In addition to this, they suggest the relationship between stock returns instead of the actual stock prices of these industries.

However, none of these provide insight into the reliable forecasting of AI company performance and, by extension, stock prices using predictive models, which is the central focus of my research. To address this gap in this line of research, my work uses SVAR and ML techniques, analyzes energy stock price index data for both traditional and renewable energy markets, and evaluates their predictions on previously unseen data to assess their effectiveness and generalizability. This use of machine learning as a modeling approach provides an insight into the AI price index in a unique way, away from more traditional econometric approaches. This would provide an alternative viewpoint to investors on the reliability of energy markets in forecasting AI markets through the lens of a predictive model, in addition to the reliance on connectedness of the volatility of these industries established through econometric models. Lastly, analyzing and comparing the importance of both non-renewable and clean energy company performances as indicators for the performance of AI companies provides insights into their individual predictive powers. This comparison was previously considered only in the context of causality, which is supplemented by this research paper.

Henriques and Sadorsky (2008) have found that shocks in the technology stock prices have a larger effect on the stock price of clean energy compared to the oil prices. In addition to this, Sadorsky (2012) also found that stock prices of clean energy firms have a higher dynamic conditional correlation with stock prices of technology firms rather than oil prices. Thus, these studies again focus on the causality and connectedness aspect of this relationship instead of a predictive one. This study aims to make contributions to the predictive branch of the energy-AI relationship through the use of forecasting and accuracy metrics, which fill in the gaps in previous studies in this area. Furthermore, this approach of predictive modeling using ML offers an additional insight into the power of energy indices combined with macroeconomic factors to forecast the AI index.

This paper discusses a detailed overview of the sources of the different price index data, along with their exploratory analysis in the succeeding section, followed by an overview of empirical methodologies that are used in this research. Eventually, this paper will portray and compare the model evaluation metrics in the results section, following a final discussion of the implications of this paper and any limitations that follow.

2. Data

This section introduces the variables and their sources which are used for analysis in this paper. In addition to this, it provides a detailed description of the dataset and an overview of the transformations on raw data to achieve the eventual cleaned dataset in use. Moreover, this section discusses the potential limitations and solutions to these limitations of the data. Lastly, a brief discussion and interpretation of summary statistics in the form of a table and a graph is mentioned to strengthen the motivation in analyzing the potential trends seen in the data.

2.1. Data overview

The data used in this research comprises reliable segment price indices for the Artificial Intelligence, Oil and Gas, and Clean Energy sectors in the United States. In addition to these indices, I include the S&P 500 index and the Effective Federal Funds Rate in order to capture any other fluctuations in the market. This dataset is of time-series form and spans from January 1st, 2021, to December 31st, 2025, recording observations at a daily frequency. However, as the stock market is closed on weekends and other national holidays, there are slightly fewer observations than that. These days are excluded to maintain continuity and a new time variable is generated based on the remaining trading days. In addition to this, there are 5 key variables along with the date of each observation. All the indices in my dataset are measured as points; however, the federal funds rate is expressed as a percentage, and the S&P 500 market Index is in USD (United States Dollar).

Based on prior research in this avenue, this paper utilizes an array of variables for analysis whose descriptions follow. First, the dependent variable that I use in this research is the NASDAQ CTA US AI and Robotics index (NQROBOUS) (Urom, Ndubuisi, Mzoughi et al. 2024). This index serves as a proxy for the performance of all U.S.-listed companies that are involved in the Artificial Intelligence and Robotics segments of technological, industrial, and other sectors (NASDAQ 2026d). Second, for one of my independent variables, I use the Dow Jones US Oil and Gas Index (DJUSEN), which is a valuable proxy for the performance of companies in the oil and gas industry in the US (S&P Global 2026). Third, I use the

NASDAQ Clean Edge Green Energy Index (CELS) as an indicator for the performance of companies that are primary manufacturers or developers in the clean energy technologies sector (NASDAQ 2026c) as my other primary independent variable. Both these energy indices serve as predictors in an attempt to forecast the stock performance of Artificial Intelligence companies. In addition to my primary variables, I include macroeconomic factors like the Federal Funds Rate (FRED 2026) as well as the S&P 500 market index (S&P Global 2025), as they relate closely to these sector indices and capture any external variation. Moreover, omitting these variables would lead to incorrect and potentially biased results.

Historical data for NQROBOUS and CELS market indices were obtained from the official NASDAQ database (NASDAQ 2026b;NASDAQ 2026a), whereas the DJUSEN index and the S&P 500 market index were obtained from Investing.com (Investing.com 2026a; Investing.com 2026b), which offers reliable market index data. Lastly, the Federal Funds Rate was obtained through the Federal Reserve Economic Database (FRED) (FRED 2026). Since the stock market suffered a crash in 2020 and did not return to its pre-crash level till late 2020, collecting data from 2021 helps to avoid any outlier values or discontinuous values from when the stock market was completely shut down. Since the range of data is quite narrow, this paper plans to utilize data of daily frequency to sufficiently extract meaningful information from the results of evaluating the econometric and machine learning models.

The CELS index consists of companies from countries other than the US, which is a potential limitation of my data; however, the majority of companies that form this index are based in the US, which makes it a reasonable proxy. Furthermore, another potential limitation of my data is due to the meeting-based updates of the federal funds rate, which leads to constant values for a given month when using it as a control variable. This is overcome through using the Daily Effective Federal Funds Rate (DFF) instead, which updates in response to macroeconomic conditions in the US.

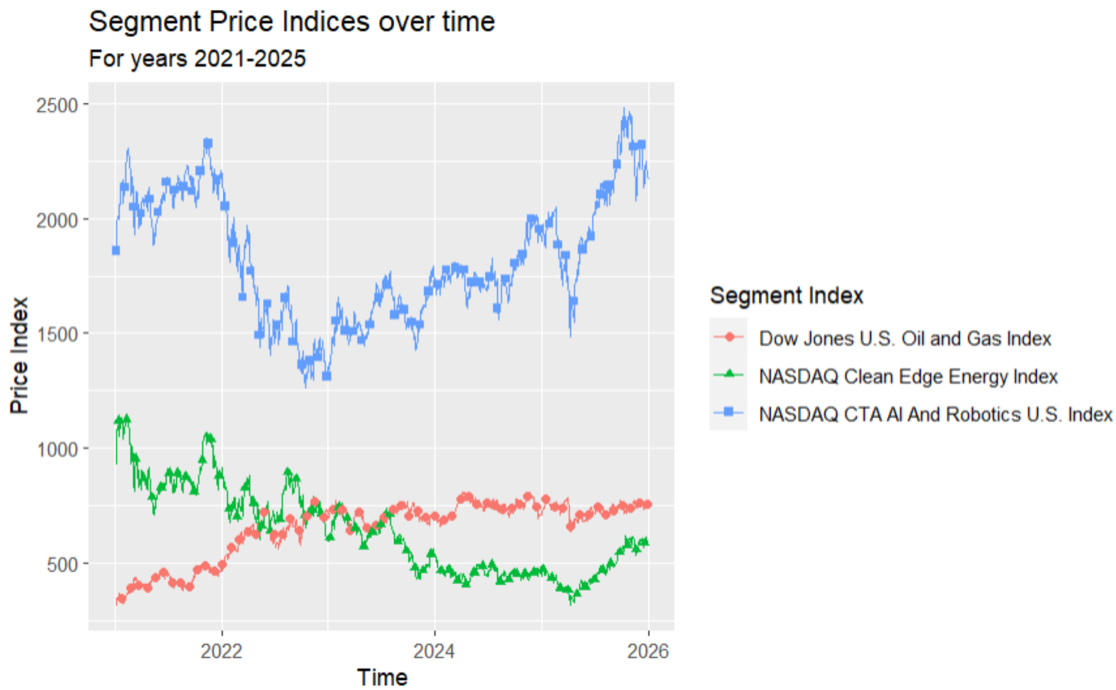

FIGURE 1. Segment Price Index (in Points) Trends from 2021 to 2025

Figure 1 demonstrates the trends of the different segment indices over time from 2021 to 2025. There has been an overall decrease in the CELS index and an overall increase in the DJUSEN index. This indicates a tradeoff between the stock prices and, in turn, the expected performances of companies in the Oil and Gas sector compared to the renewable energy sector. Furthermore, the decrease in CELS index was accompanied by a decrease in NQROBOUS index up until 2023, which might show some relationship between these indices; however, the trends are opposite post 2023. Alternatively, the DJUSEN index is slightly increasing and becomes steady after 2023, which suggests that it might have a lagged effect on the increase in the NQROBOUS index, which occurs after 2023. However, these co-movements are not easily evident, which prompts the need for statistical analysis in identifying the predictive power of energy indices in forecasting the performance of AI stocks.

TABLE 1. Average Market Index levels over 5 years

Year	DFF*	CELS	NQROBOUS	DJUSEN	S&P 500
2021	0.079	907.890	2123.117	425.051	4273.408
	(0.013)	(99.235)	(95.649)	(40.552)	(287.460)
2022	1.696	752.521	1603.842	649.345	4098.515
	(1.415)	(68.298)	(217.506)	(67.966)	(291.831)
2023	5.028	612.355	1582.641	704.865	4283.729
	(0.338)	(82.231)	(92.799)	(33.261)	(230.258)
2024	5.144	456.401	1768.214	745.902	5428.235
	(0.309)	(22.792)	(97.052)	(34.242)	(368.071)
2025	4.211	472.789	2042.841	733.683	6216.866
	(0.207)	(82.031)	(224.816)	(29.162)	(462.406)

*NQROBOUS = NASDAQ CTA US AI and Robotics Index, DJUSEN = Dow Jones US Oil and Gas Index, CELS = NASDAQ Clean Edge Green Energy Index, DFF = Effective Federal Funds Rate

Standard deviations in parentheses

Table 1 conveys information consistent with Figure 1 as the average CELS index decreased sharply every year till 2024 from 907.890 in 2021 to 456.401, followed by a slight increase to 472.789 in 2025. This decrease is similar to the NQROBOUS index, which decreased from 2123.117 in 2021 to 1582.641 in 2023, but was followed by an increase in the NQROBOUS index back to 2042.841, which is close to the 2021 level, which suggests some co-movement of these indices, as also supported by Figure 1 discussed above. This increase in the NQROBOUS index is again consistent with the increase in the DJUSEN index from 2021 to 2023, which suggests a lagged effect on the NQROBOUS index. The DJUSEN index increased from 425.051 in 2021 to 745.902 in 2024, but decreased slightly to 733.683 in 2025, showing a tradeoff with the CELS index, which increased in 2025. Furthermore, the S&P 500 index decreased slightly from 4273.408 to 4098.515 in 2022 but increased thereafter to 6216.866 with sharp increases in 2024 and 2025. Similarly, the Federal Funds rate increased until 2024 but then decreased abruptly from 5.144 to 4.211 in 2025.

2.2. Data Construction

Due to the multiple data sources for my research, one of the initial steps in data preparation was generalizing the feature names in order to merge them appropriately. Due to the same reason, the date formats were widely different across my sources, which also needed standardization to enable matching while merging the various datasets. Furthermore, the data included other variables that were not necessary for analysis and modeling and were excluded.

3. Empirical Methodology

This section will introduce the workings of the SVAR model and its suitability for the research question at hand. In addition to this, it will include the assumptions that need to be satisfied prior to implementing the SVAR in order to validate the results. Furthermore, this section will introduce the Machine Learning Models that I intend to use with a detailed explanation of generating training and testing splits to evaluate the effectiveness of these models on out-of-sample data.

3.1. SVAR

3.1.1. Model and Feature overview

The SVAR model with a levels specification is the primary method in this research paper, as shocks in the energy industry might often be transmitted to the AI industry through some macroeconomic factors. Thus, this model accurately captures both the contemporaneous and delayed effects of shocks in the energy indices on the AI index, thus making it a suitable model in this case. Furthermore, because these variables have varied contemporaneous effects on one another, an ordered structural model might better capture their relationships than other linear models. This paper imposes a short-run restriction on the SVAR model to accurately capture these differences in contemporaneous effects and produce reliable results.

The estimated Structural VAR (SVAR) model can be written as:

$$AY_t = C_1 Y_{t-1} + \epsilon_t$$

where,

$$Y = \begin{bmatrix} DJUSEN_t \\ CELS_t \\ NQROBOUS_t \\ SP500_t \\ DFF_t \end{bmatrix}, \epsilon_t = \begin{bmatrix} e_{DJUSEN,t} \\ e_{CELS,t} \\ e_{NQROBOUS,t} \\ e_{SP500,t} \\ e_{DFF,t} \end{bmatrix},$$

A is a lower triangular matrix and C_1 is the matrix of coefficients for the lagged terms. The lag order of 1 was chosen based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) using the training data.

The primary variable of interest in this model is the NQROBOUS Index which serves as a proxy for performance of US companies involved in Artificial Intelligence and Robotics.

The secondary variables include the NASDAQ Clean Edge Energy Index (CELS), Dow Jones US Oil and Gas Index (DJUSEN), S&P 500 Global Index (SP500), as well as the effective federal funds rate (DFF). These indices serve as market capitalization weighted aggregates across the stocks of publicly traded US companies in the given sector thus serving as effective proxies for the company performance. Since the SVAR model determines all variables simultaneously, there is no clear dependent or independent variable. However, the primary goal in the scope of this research is to identify the dynamic relationship between these secondary variables and the NQROBOUS Index for AI companies in the United States.

3.1.2. Identification Assumptions

The identification strategy in this case is the specific ordering of the variables in the equation according to a decreasing order of exogeneity to achieve the Cholesky recursive structure using short-run restrictions. The DJUSEN index is treated as the most exogenous variable in this model, followed by the CELS index, the NQROBOUS index, the S&P 500 index, and lastly the Effective Federal Funds Rate.

This ordering is assumed according to economic theory, as the DJUSEN index is primarily affected due to shocks in oil prices, which often occur due to geopolitical tensions or output cuts by major producers, leading to oil supply shocks (Kilian 2009). In addition to this, it is a traditional energy source, and thus it serves as a substitute for clean energy. Due to this, shocks in the DJUSEN index would contemporaneously affect the CELS index as well. Furthermore, previous research has found that clean energy markets are affected by oil and gas markets due to a spillover effect (Du and Xu 2024). Thus, this ordering of the DJUSEN index as the most exogenous variable, followed by the CELS index, is a valid assumption for this research. This is also evidently seen in the index trends in Figure 1, where the decrease in the CELS Index is accompanied by a corresponding increase in the DJUSEN Index. Furthermore, the AI index is placed next in the ordering of the SVAR model. Assuming that breakthroughs in the AI sector often involve new energy inputs, either renewable or non-renewable, this AI index thus changes as a result of shocks in the energy indices. Also, the pivotal role of energy in sustaining AI as a raw material in its production, as mentioned earlier in the paper, further suggests the validity of this ordering assumption in the paper (IEA 2025). Thus, since AI is directly dependent on energy, energy stocks are likely to have a contemporaneous effect on the AI index.

Since the S&P 500 is aggregated across a multitude of sectors, it takes into account these changes in the energy and AI indices as they form a considerable component of this index. Thus, these indices together have a contemporaneous effect on the S&P 500 index. Additionally, neither of these energy indices would contemporaneously respond to the S&P 500 index for this reason. Furthermore, macroeconomic factors like the effective

Federal Funds Rate may adjust according to the performance of the stock market to maintain financial stability. Thus, this change in all the energy indices, and in turn the S&P 500 index, would cause changes in the level of the Effective Federal Funds Rate through changing macroeconomic conditions. Hence, the assumption ordering of the variables is appropriate and is consistent with economic theory and past literature for this research.

Some possible threats to identification include the presence of features that act as intermediaries in this chain of effects described above. The inclusion of the S&P 500 index and Federal Funds rate serves to minimize the effects of these excluded variables as they capture the overall macroeconomic and equity market conditions effectively while accounting for the omitted variable bias. Another threat includes the wrongful ordering of variables, which would then void the causality aspect of this research. All combinations of the energy and AI indices are modeled as robustness checks to assess the sensitivity of the results to this ordering assumption. Lastly, other threats include using a conservative number of lags in this Energy-AI relationship, which would not correctly capture the delayed effects of the features on each other and, in turn, lead to biased coefficients. Models with additional lags are also computed as additional robustness checks, which are described in the following section.

3.1.3. Robustness Checks

Some key robustness checks include switching the ordering of the indices in the SVAR model to judge the sensitivity of results based on this ordering in the research, which are discussed in the results section. Further robustness checks involve excluding the Federal Funds Rate and S&P 500 index in alternative specifications to justify their importance in the energy-AI relationship in the original model. These sets of checks would test the use of macroeconomic indicators to judge the sensitivity of the effects of energy indices on the AI index. The responses of the AI index to shocks in the energy indices have similar trends without these macroeconomic indicators, as seen in the Appendix. Lastly, some additional robustness checks would include models with a greater number of lagged terms, which would help to test the robustness of this SVAR model to the lag structure. As seen in the figures in the appendix, the SVAR model results are robust to the changes in the lag order. This is evident from the similar trends in responses of the AI index to shocks in the secondary variables in the model. Though the 95% confidence intervals are much wider, the significance remains, suggesting the robustness of the results to the degree of lags used in this model.

3.2. Machine Learning with SVAR foundation

In order to compare the results from the SVAR model, this research utilizes the inputs to the SVAR framework as features to be input into a Machine Learning Model. This research uses Support Vector Regression (SVR) and Extreme Gradient Boosting Regression (XGBoost) as models for comparison, as they are standard models in this avenue of research and are used for comparison in established papers like Oukhouya and El Himdi (2023) and Alhomadi (2021).

The input features to these models are identical to the SVAR model and thus include all variables with a lag order of 1 in addition to the CELS index and DJUSEN index, which were assumed to have a contemporaneous effect on the AI Index. The target variable in these models is the NQROBOUS index. The XGBoost model helps to identify non-linear trends in the data with the help of a tree-based model, which uses the residuals of one tree to train the next tree, thus eventually leading to a valuable end model. Thus, this model offers an alternative method to identify trends and improve prediction accuracy through non-linearity. The Support Vector Regression uses a kernel function to make predictions thus capturing the trends not previously captured by the linear or tree-based models. These ML models don't have any specific identification assumptions as they are primarily big data tools that are used for prediction purposes.

Additionally, since the SVR is a distance-based model and there is a clear difference in the scale of variables, the training data and testing data were first scaled before being used in the model to ensure optimal performance. However, XGBoost was fit to the non-scaled features due to the tree-based decision making involved in this model, which does not depend on the scale of the features in the model.

3.3. Feature Engineered Machine Learning Models

To serve as a benchmark for prediction accuracy, this research further involves training ML models using other generated features in addition to the SVAR features. The primary dependent variable remains the same; however, the predictors were modified to include the first three lags of all indices and the effective federal funds rate to capture the most recent trends in stock prices. In addition to this selection of lags, technical indicators for all indices were also used to capture the momentum in the AI index, which is not already evident from the lagged price indices. These indicators, like the Relative Strength Index (RSI), Simple Moving Average (SMA), and Exponential Moving Average, were inspired by published research like Agrawal, Khan, and Shukla (2019), who used them to predict index prices of stocks. These technical indicators were calculated over a period of 10 days to capture the most recent information. The ML models used in this branch are the same as the ones mentioned above and thus do not require testing for any identification restrictions.

Both these types of ML models require tuning of hyperparameters within them, which involves an exhaustive search of various permutations of the model parameters to identify the optimal combination. This search is conducted through repeated training of models on increasingly larger subsets of the training data while evaluating them on held-out observations. The parameter combination that offers the greatest predictive power is then selected for use in the final model, which is subsequently evaluated on out-of-sample observations to assess generalizability. This process helps to maximize the predictive power of these models.

3.4. Training-testing split

To truly understand the effectiveness of a certain predictive model, there is a need for testing with out-of-sample data. This paper involves a train-test split, where all models are trained on data spanning from 2021 through 2024, and the 2025 observations are used as a benchmark for predictions and for testing prediction accuracy. The primary metrics of prediction accuracy used in this paper include the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE). These results are then compared across all sets of models.

4. Results

This section will discuss the coefficients obtained through the models, along with their significance. In addition to this, it will include accuracy metrics, like root mean squared errors and mean absolute errors, which portray the effectiveness of both the trained models on previously unseen data. Furthermore, it will discuss the comparison of the two modeling approaches to determine the strengths and weaknesses of each approach. Lastly, it will include an appropriate discussion of forecasting for either model through the use of informative graphics demonstrating the predictions.

4.1. SVAR Impulse Responses

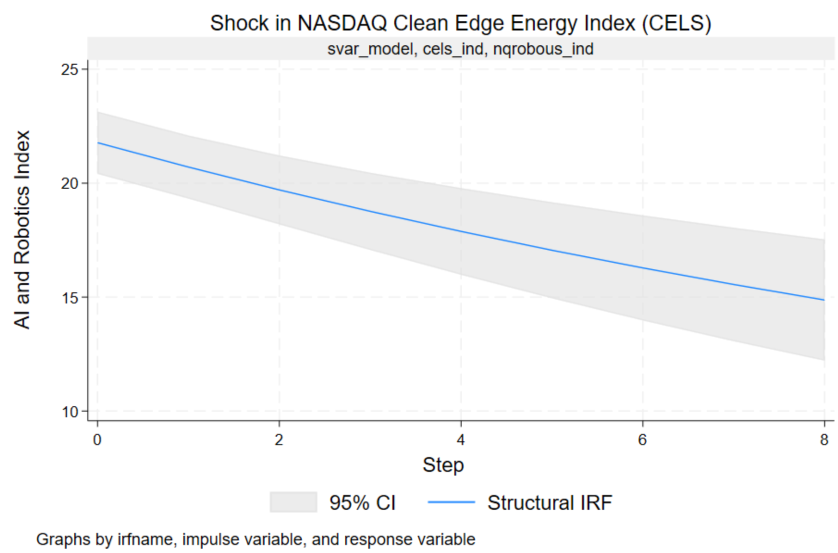

FIGURE 2. Impulse Response Function for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in NASDAQ Clean Edge Energy Index (CELS)

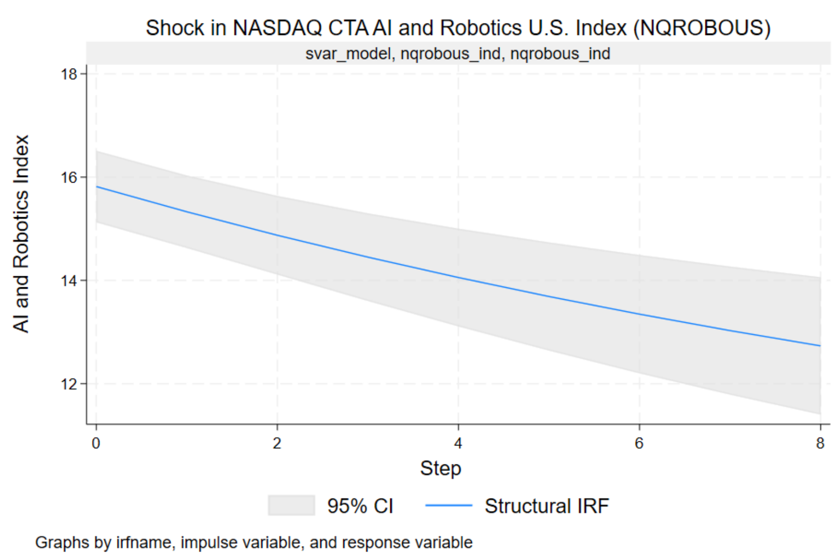

FIGURE 3. Impulse Response Function for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in itself

For the SVAR model computed in this paper, a one-unit shock to the Clean Energy Index appears to have a significantly large immediate and long-term effect on the AI Index, as seen in Figure 2. This suggests that the CELS index has significant dynamic effects on the NQROBOUS index, with a larger magnitude than a shock in the AI index itself (Figure 3).

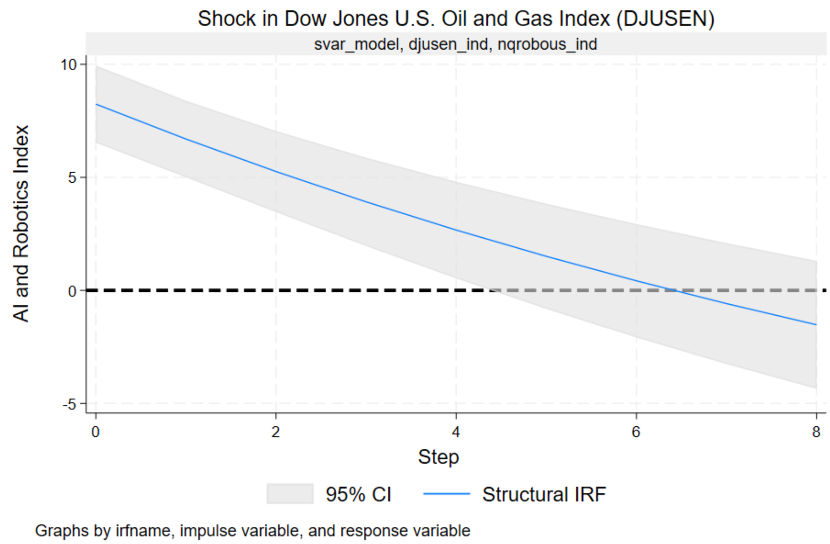

FIGURE 4. Impulse Response Function for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in Dow Jones U.S. Oil and Gas Index (DJUSEN)

Furthermore, a one-unit shock to the Oil and Gas index has a moderate immediate positive impact on the AI index; however, in the long run, the effect of this shock moves towards insignificance. This response of NQROBOUS to a shock in DJUSEN is much smaller in magnitude than the response to the Clean Energy index, as seen in Figures 2 and 4, suggesting the greater power of clean energy shocks in changing AI stock prices compared to more traditional sources of energy. Thus, shocks in this index can potentially contain information about AI stock prices in tandem with the Clean Energy Index, as shocks to both of them lead to significant positive effects on the AI Index. Moreover, shocks to the effective federal funds rate and the S&P 500 Index do not appear to have a significant impact on the AI Index, which can be seen in the Impulse Response Functions in the Appendix.

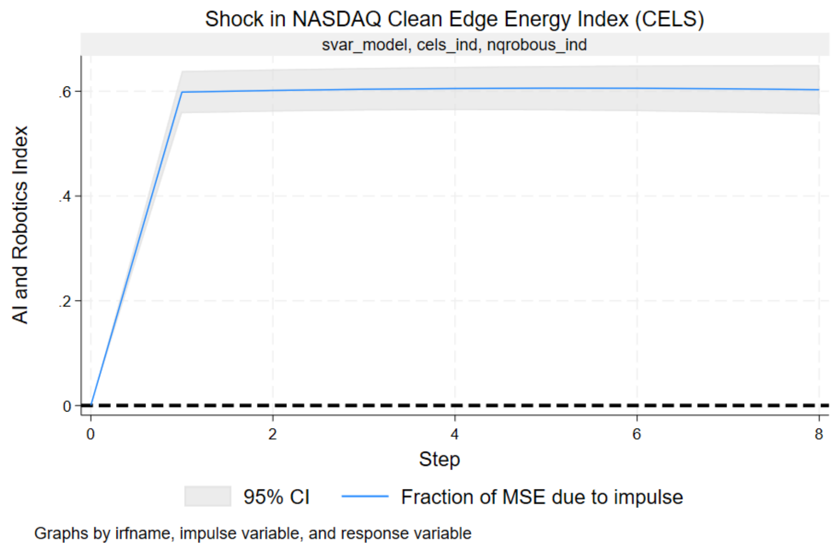

FIGURE 5. Forecast Error Variance Decomposition (FEVD) for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in NASDAQ Clean Edge Energy Index (CELS)

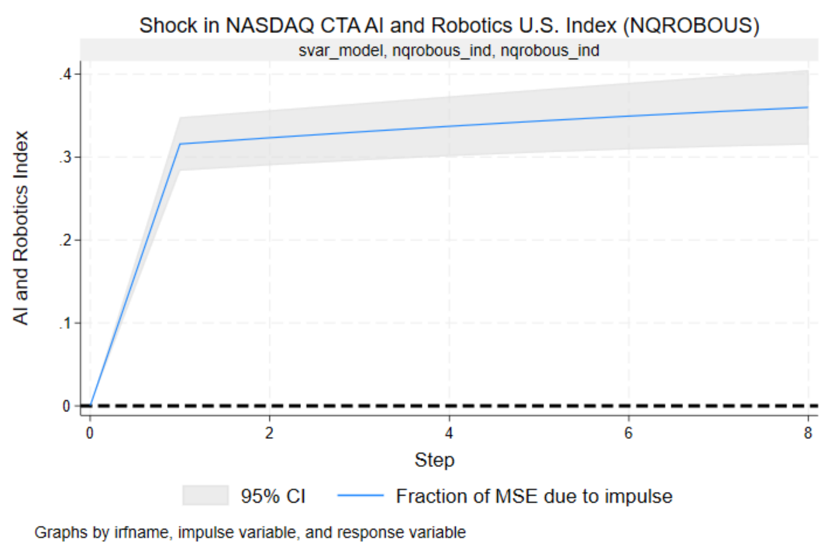

FIGURE 6. Forecast Error Variance Decomposition (FEVD) for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in itself

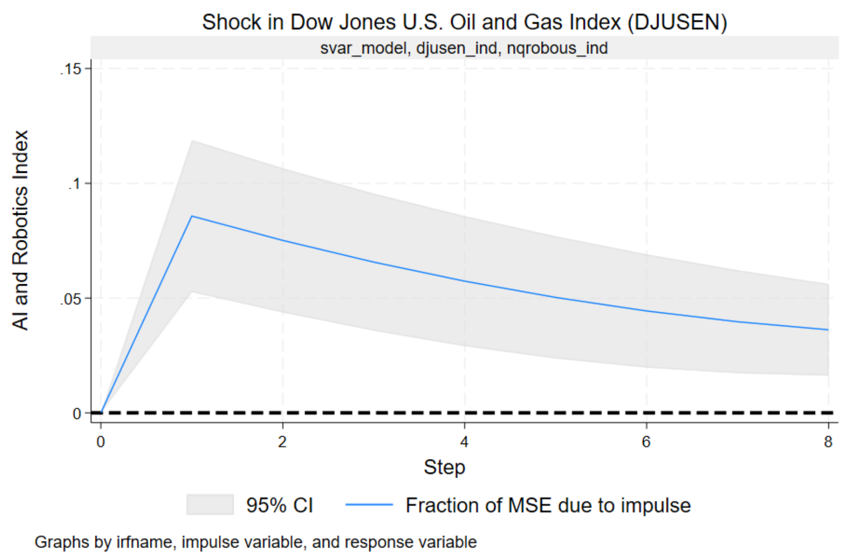

FIGURE 7. Forecast Error Variance Decomposition (FEVD) for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in Dow Jones U.S. Oil and Gas Index (DJUSEN)

Shifting the focus to forecasting, an impulse in the Clean Energy Index is also a major component of the forecast error variance in the AI Index (around 60%). This again stresses the importance of clean energy stock price shocks in explaining the variance of forecasting errors in the AI company stock performances. Impulse in the NQROBOUS index is a slightly smaller component of this error in forecasting(around 33%). Lastly, the DJUSEN index forms an even smaller component of this error (around 7%), as suggested by Figures 7,8, and 9. The S&P 500 index and the Effective Federal Funds Rate contribute negligibly to this forecast error without any significance as evidently seen in the figures in the appendix.

These graphs together indicate that a shock in the stock prices of the Clean Energy Companies can serve as a good insight for investors and policymakers to expect and prepare for a potential change in the stock performance of AI companies as well. Furthermore, this impact provides more information about the AI Index than its own lagged value, which is often a good estimate of the next time period. However, based on the results from the robustness checks, these results and interpretations, specific to the Clean Energy Index, are sensitive to the ordering assumption for the SVAR model.

Based on a robustness check in Figure 8, when the ordering of the variables in the SVAR model is modified, it appears that the results are sensitive to the ordering assumption. This is because the effect of a shock in the CELS index on the AI index is significant only when it is ordered before the AI index, which suggests the contemporaneous nature of this relationship. Moreover, this response in NQROBOUS due to its own shocks increases when the CELS index is ordered after this index, which suggests the absorbing nature of this shock response in the absence of a contemporaneous effect of the CELS index (Figure

9). Furthermore, the DJUSEN index shocks induce a significant response in NQROBOUS in all orderings. However, the direction of impact of this shock response varies across orderings, as can be seen in Figure 10.

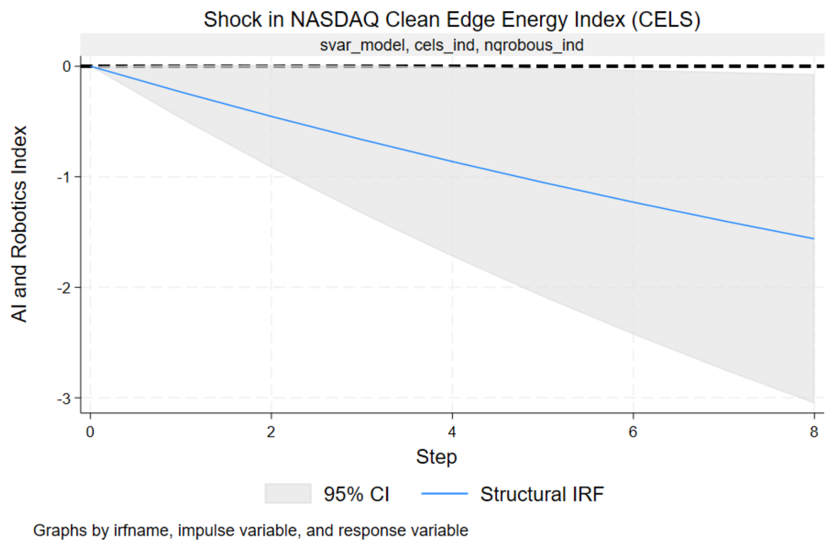

FIGURE 8. Impulse Response Function for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in NASDAQ Clean Edge Energy Index (CELS) for an alternate ordering of variable -> AI Index, Oil and Gas Index, Clean Energy Index, S&P 500 Index, and Effective Federal Fund Rate

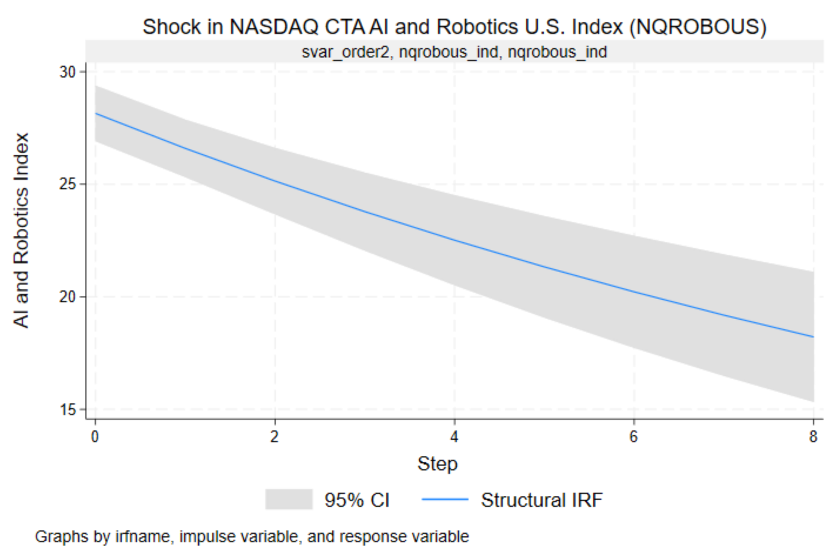

FIGURE 9. Impulse Response Function for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) on itself for an alternate ordering of variable -> AI Index, Oil and Gas Index, Clean Energy Index, S&P 500 Index, and Effective Federal Fund Rate

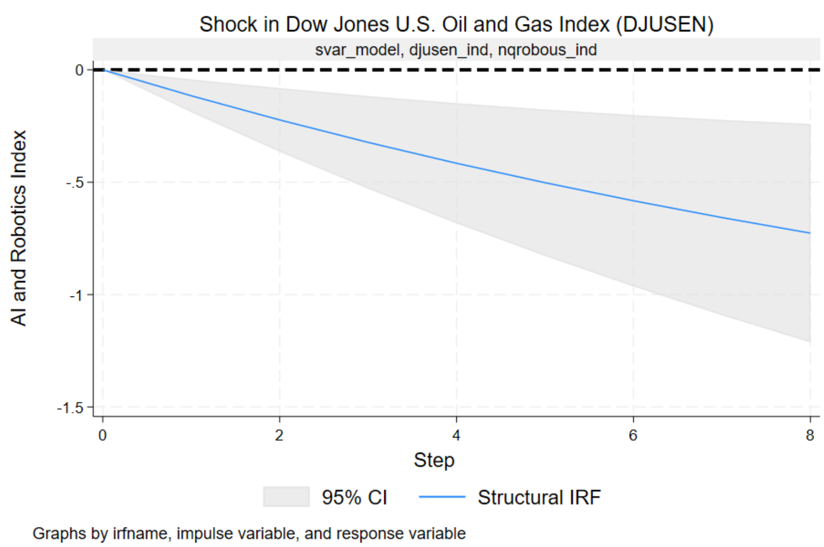

FIGURE 10. Impulse Response Function for NASDAQ CTA AI and Robotics U.S. Index (NQROBOUS) for a shock in Dow Jones U.S. Oil and Gas Index (DJUSEN) for an alternate ordering of variable -> AI Index, Oil and Gas Index, Clean Energy Index, S&P 500 Index, and Effective Federal Fund Rate

4.2. Forecasting Errors across all Models

This section compares the prediction errors for different modeling approaches.

The SVAR model performs poorly in forecasting values of the AI price index on out-of-sample data, as it doesn't accurately capture the volatility in observations, unlike the other models. This is partly due to the dynamic forecasting nature of the SVAR model, where the errors of one prediction accumulate over time, compared to the direct predictions in the ML models. The ML models with the same inputs as the SVAR model perform significantly better, perhaps due to capturing the non-linear trends evident in the price index of AI. The worst-performing ML model still has almost a 300 unit improvement in the Root Mean Squared Error and Mean Absolute Error over the SVAR model with the same inputs.

Additionally, the SVR model with technical indicators and additional lags performs much better than the baseline SVR model, which suggests the valuable information added to the model by these indicators, as the errors reduce by a large amount. This improvement is not evident across XGBoost models. These results suggest the significant predictive power of momentum-based features in SVR when attempting to predict the AI price index, which is especially volatile.

Lastly, both the Support Vector Regression (SVR) models perform significantly better than the Gradient Boosting models, signaling the better suitability of the kernel-based function in capturing the trends in the AI index. This is evident from Table 2 below, as the best performing ML model is the Support Vector Regression with technical indicators, with a Root Mean Squared Error of only 16 units, which is negligible as the average of this index is around 1800 units. The Support Vector Regression model is a valuable tool for investors to get some insight into the price of AI stocks, which would play the role of one of the important factors in trading.

Hence, the ML models offer useful information for investors on the stock price index of AI companies in the future, whereas the SVAR model provides policymakers with valuable insights about the effect of clean energy and non-renewable energy shocks on the AI sector. Thus, this research aids both these parties in making decisions in their respective fields.

TABLE 2. Model Performance Comparison on Out-of-sample data

Model	**RMSE**	**MAE**
SVAR	378.945	305.227
SVR (Baseline)	25.606	19.440
XGBoost (Baseline)	67.977	46.051
SVR (Generated Features)	15.815	11.696
XGBoost (Generated Features)	67.364	44.400

This difference in forecasting is evident from Figures 11 and 12 as the SVR model accurately captures the direction of movements of this AI index compared to the SVAR model. The SVAR model essentially predicts a smooth trend, which is often unrealistic in financial data such as stock indices. This smooth prediction is primarily due to the model's linearity, which produces underwhelming results given the persistent non-linearities in stock data, in addition to the dynamic forecasting nature of this model. In contrast, the SVR model, which uses a kernel-based function, captures this volatility in the AI index appropriately overall, as evident from the figure above. These forecasting visuals reiterate the out-of-sample predictive power of big data tools like Machine Learning in predicting the AI price index compared to structural econometric models like the Structural Vector Auto-Regression Model.

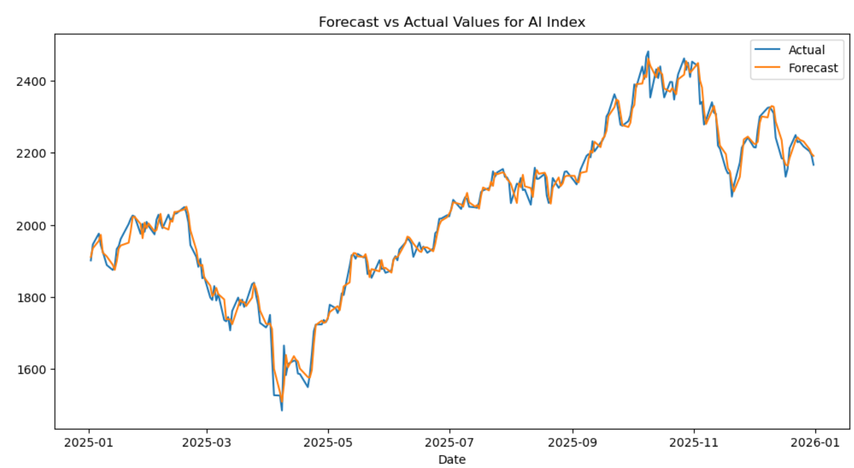

FIGURE 11. Out-of-Sample SVR Forecast for NASDAQ CTA Artificial Intelligence and Robotics U.S. Index (NQROBOUS)

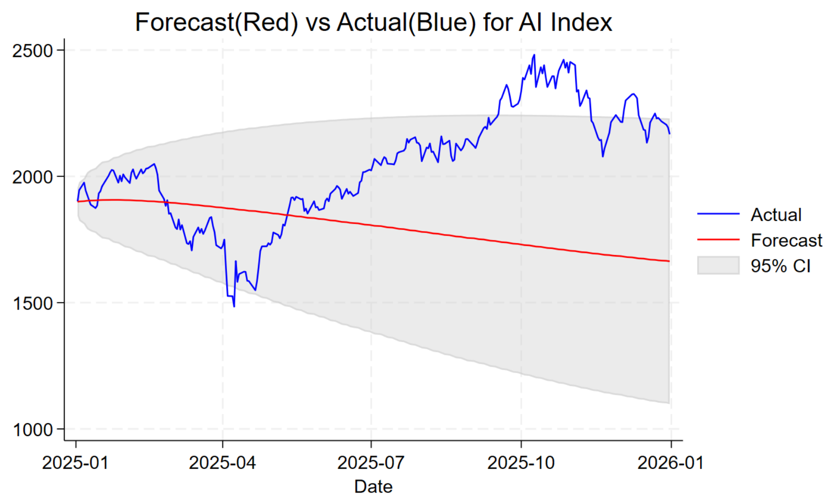

FIGURE 12. Out-of-Sample SVAR Forecast for NASDAQ CTA Artificial Intelligence and Robotics U.S. Index (NQROBOUS)

5. Conclusion

This section highlights breakthroughs and contributions that are made by this paper in the form of a summary of the primary results. Furthermore, it will discuss the potential consequences of these findings on policies as well as investor sentiments. Lastly, it will mention the potential limitations of this research and point out some future avenues for research, which might be an interesting supplement to this paper.

5.1. Summary and Implications

This paper essentially attempts to evaluate the predictive power of energy indices like the NASDAQ Clean Edge Energy Index and the Dow Jones U.S. Oil and Gas Index in forecasting values of the NASDAQ CTA AI and Robotics U.S. Index. This paper presents some major insights into the structural relationship between these indices through Impulse Response Functions and Forecast Error Variance Decompositions, as well as the predictive accuracy of these methods through metrics like RMSE and MAE. The shocks in the Clean Energy Index have a significant positive impact on the AI Index for the specific ordering assumption used in the SVAR model. However, this relationship is insignificant for orderings where the Clean Energy Index is assumed not to have a contemporaneous effect on the AI Index. Furthermore, shocks in the Oil and Gas Index significantly affect the AI Index for all specifications of the short-run restriction SVAR model, thus showing a more robust relationship compared to the Clean Energy Index. Lastly, ML models like Support Vector Regression and Gradient Boosting Regression offer enhanced predictions relative to econometric models like SVAR, due to capturing the true trend of the AI Index better. Furthermore, the Support Vector Regression's superior relative performance to Gradient Boosting suggests that kernel-based functions, rather than tree-based boosting methods, identify this trend more effectively.

These results are useful for policymakers, as knowing the significant relationship between Oil and Gas index shocks and the AI index helps to identify any potential risks to the financial stability of the AI sector. This can be done through supervision and industry-specific policies in an attempt to influence the performance of stocks in the traditional energy market. Furthermore, from an investor's point of view, the satisfactory performance of the Support Vector Regression Model means that they provide useful predictive insights, which can be taken into account when making trading choices in addition to other factors.

5.2. Limitations and Future Work

As the results show, a potential limitation of this study is the sensitivity of the AI Index's response to Clean Energy Index shocks to the relative order in the identification assumption, which affects the interpretation of these results. Furthermore, this paper uses standard Machine Learning Models, which are significantly less complex than Artificial Neural Networks like Long Short-Term Memory (LSTM) Models (Hochreiter and Schmidhuber 1997) that might offer better predictive performance.

Some future work to add to this paper would be an analysis of feature importance, which is how much each feature contributes to the prediction in a model, of the inputs to the Machine Learning Models, and a comparison with the SVAR results. This would provide a deeper insight as to how these features contribute to each model. In addition to this, using more complicated machine learning models like Neural Networks may boost the accuracy on out-of-sample observations.

Another potential avenue of research would be to change the scope of this study to include a particular big AI firm's stock price and how it is affected by the stock prices of its major suppliers of energy. Instead of generalizing these firms to a sector index, this approach would help to obtain insights at the level of individual firms.

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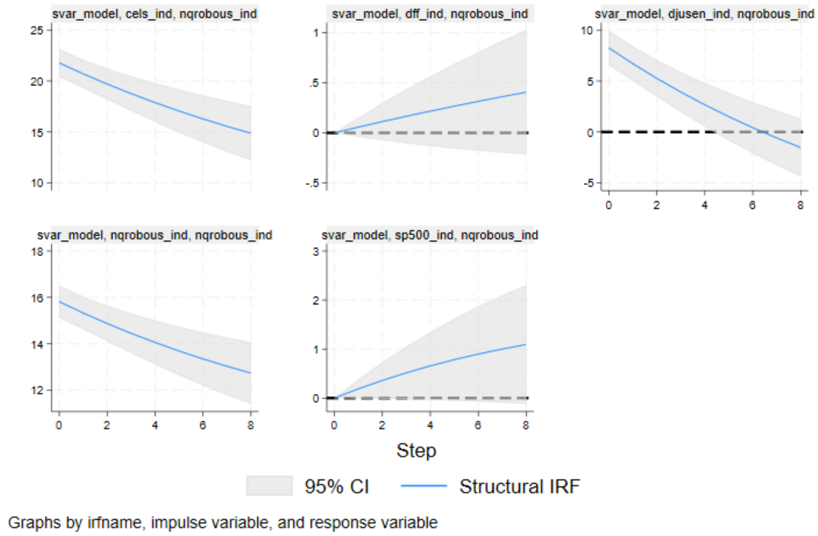

FIGURE A1. Main Model IRFs

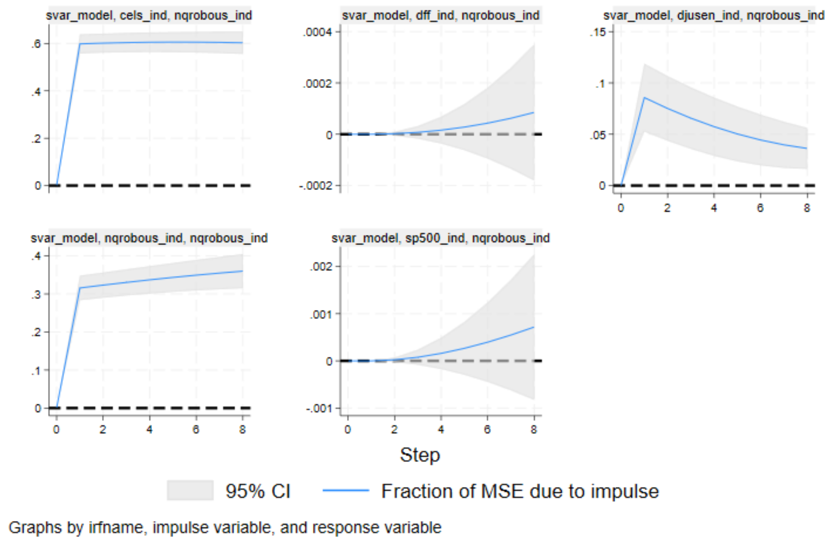

FIGURE A2. Main Model FEVDs

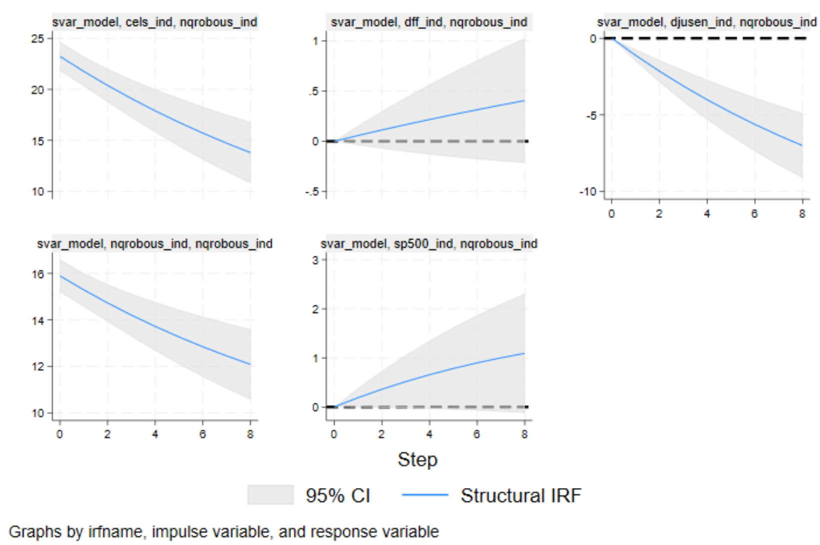

FIGURE A3. IRFs for CELS, NQROBOUS, DJUSEN, S&P500, DFF Ordering

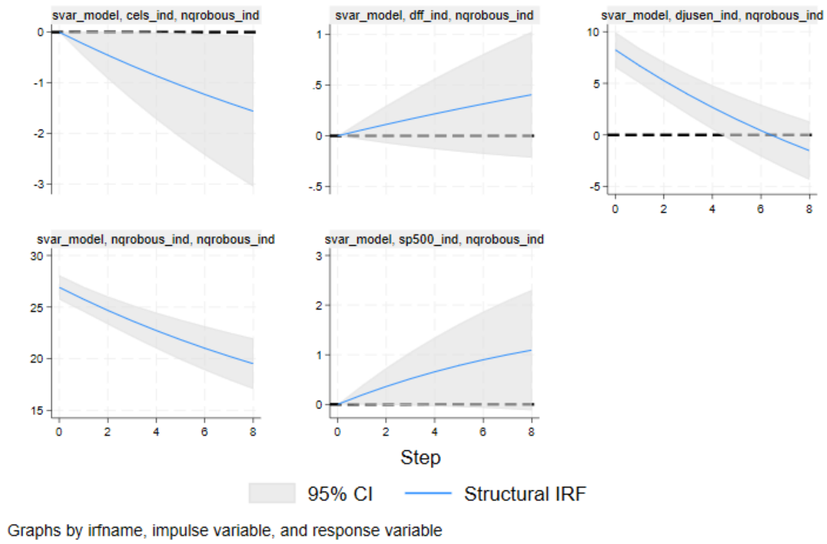

FIGURE A4. IRFs for DJUSEN, NQROBOUS, CELS, S&P500, DFF Ordering

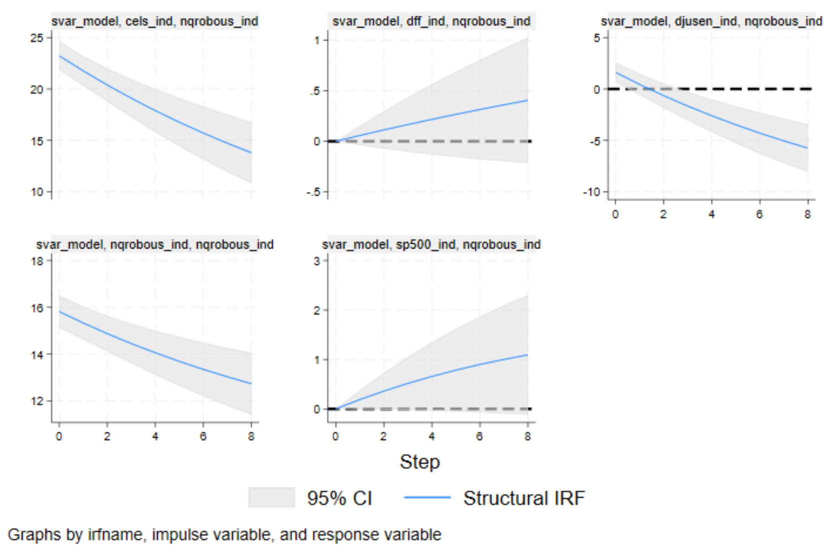

FIGURE A5. IRFs for CELS, DJUSEN, NQROBOUS, S&P500, DFF Ordering

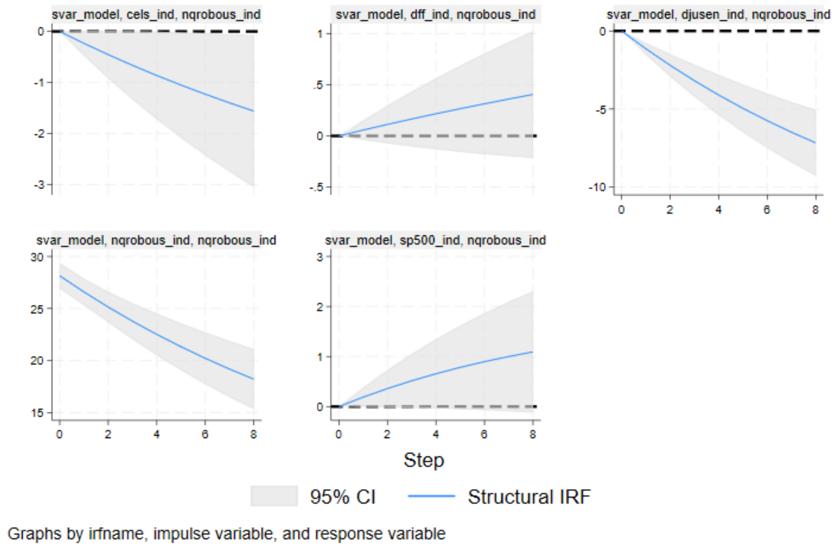

FIGURE A6. IRFs for NQROBOUS, DJUSEN, CELS, S&P500, DFF Ordering

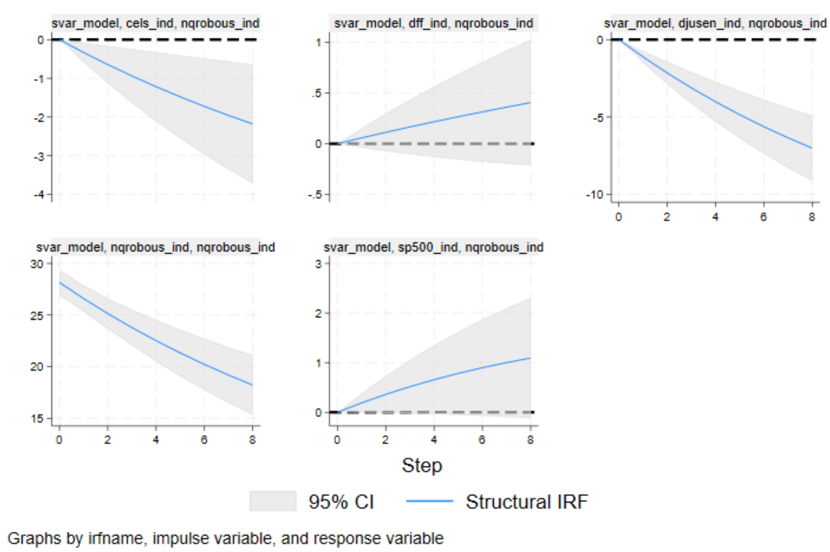

FIGURE A7. IRFs for NQROBOUS, CELS, DJUSEN, S&P500, DFF Ordering

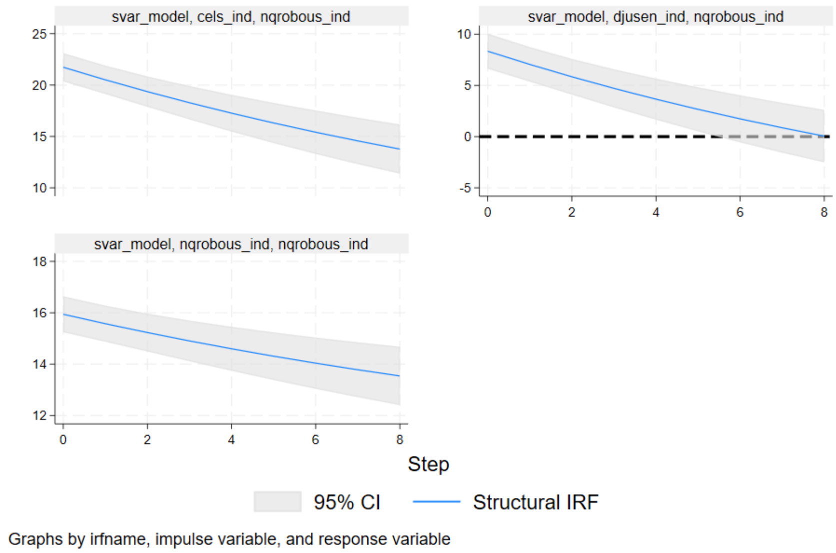

FIGURE A8. IRFs without the Effective Federal Funds Rate and S&P 500 Index

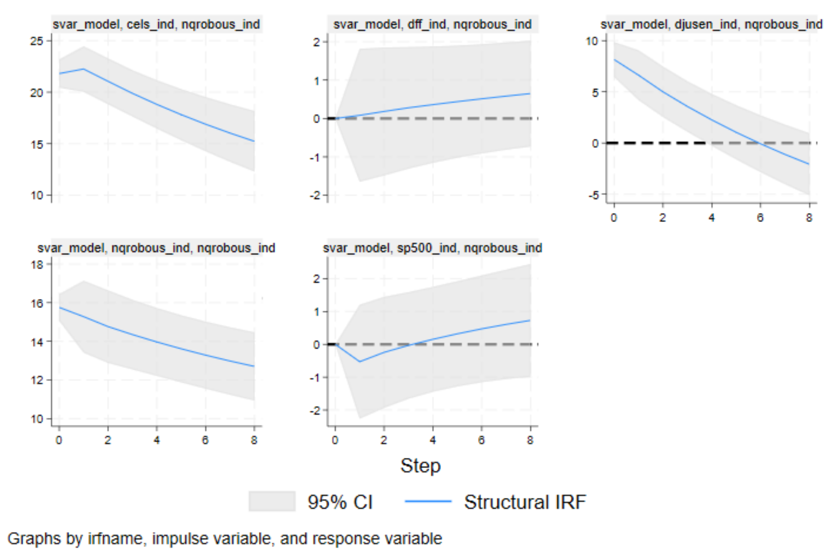

FIGURE A9. IRFs with lag order 2

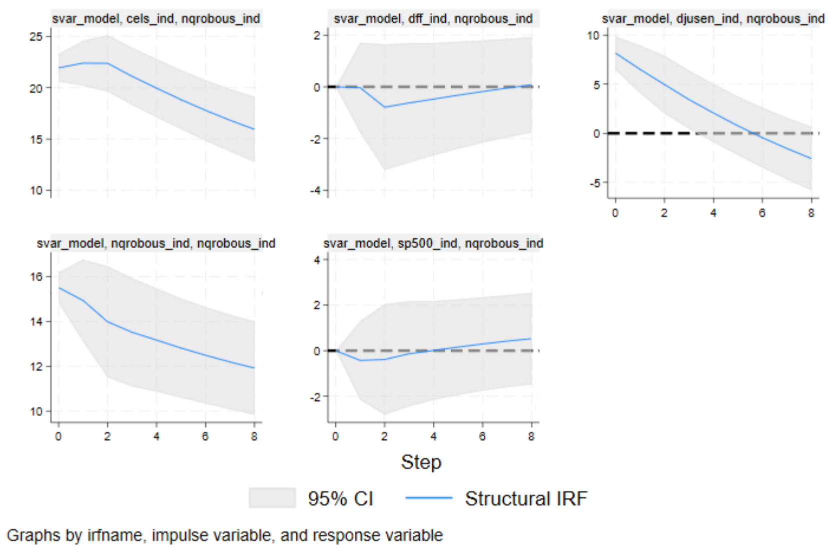

FIGURE A10. IRFs with lag order 3